Babu Banarasi Das University, Lucknow
U.P

Descriptive Analytics Report
Toyota Global Vehicle Sales Performance 2019-2025

Tracking Best-Sellers, Sales Trends, Ratings, Regional Demand and Pricing

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Date – 13-08-2026

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Descriptive Analytics Report : Daily Purchase Behavior at Scale

Note on data: Toyota's live transaction data isn't publicly accessible, so this report uses a synthetic dataset built to mirror real Automobile industry patterns **(20 categories across 4 regions, 7 years, ~300k simulated units).** It's designed to show exactly what a descriptive analytics pipeline processing millions of daily purchases would produce — the same structure applies once real data is substituted in.

Objective

1. What are best-selling cars?
2. How the cars are rated and when the rating was started slipping?
3. In which year most of the cars are sold out?
4. Which region generates the highest total unit sales, and has that ranking shifted year over year?
5. What is the average on-road price by region, and how much of that gap is due to taxes/registration vs. the base ex-showroom cost?
6. Which region is most profitable per unit sold, and which is least?
7. Is there a growing or declining trend in total sales volume from 2019 to 2025?
8. Which models have the highest average rating, and which have the lowest?
9. Which model-region combination is the single biggest revenue driver?

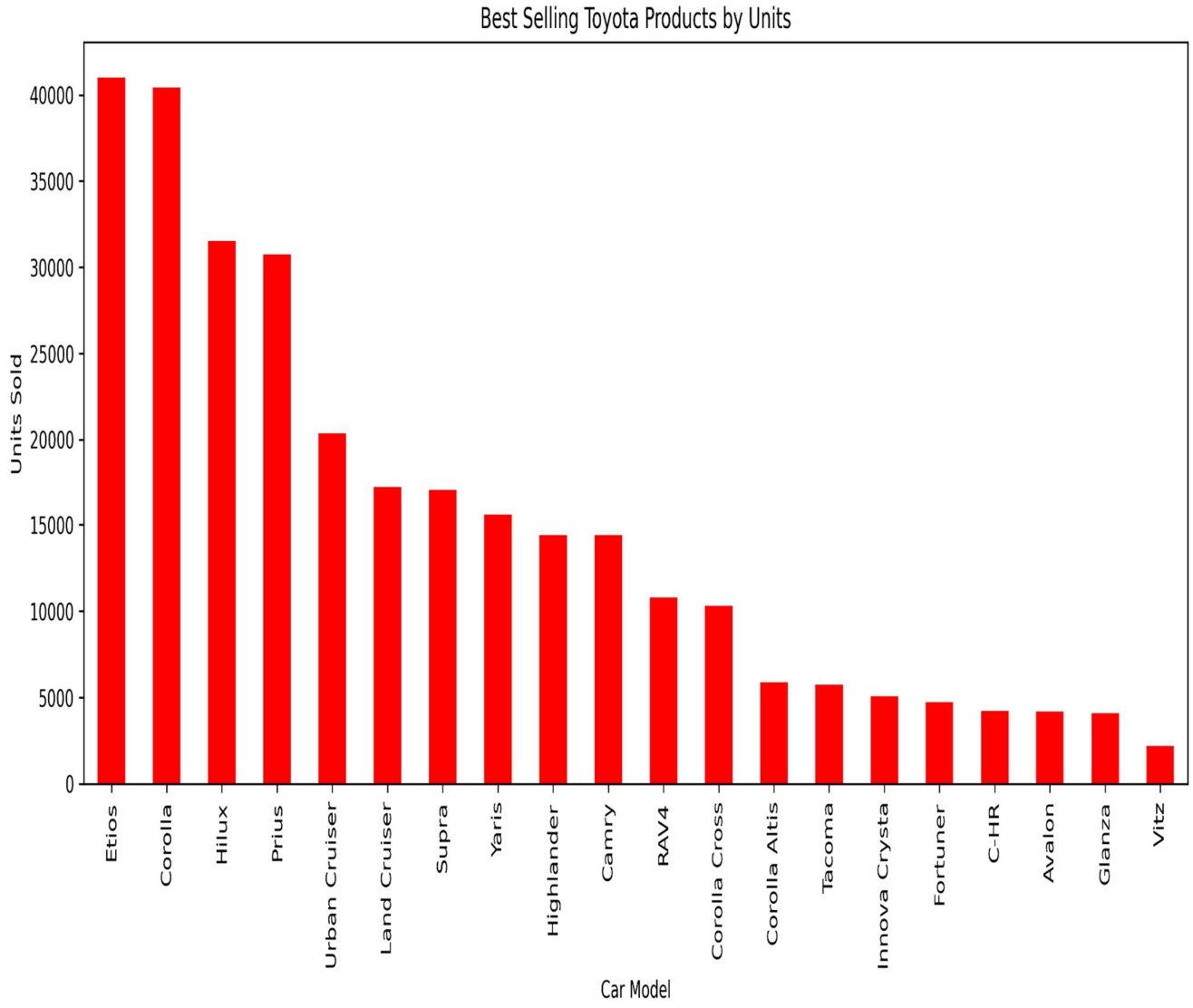
How This Would Work at Real Scale

Processing "millions of purchases daily" is an engineering problem before it's an analytics one.

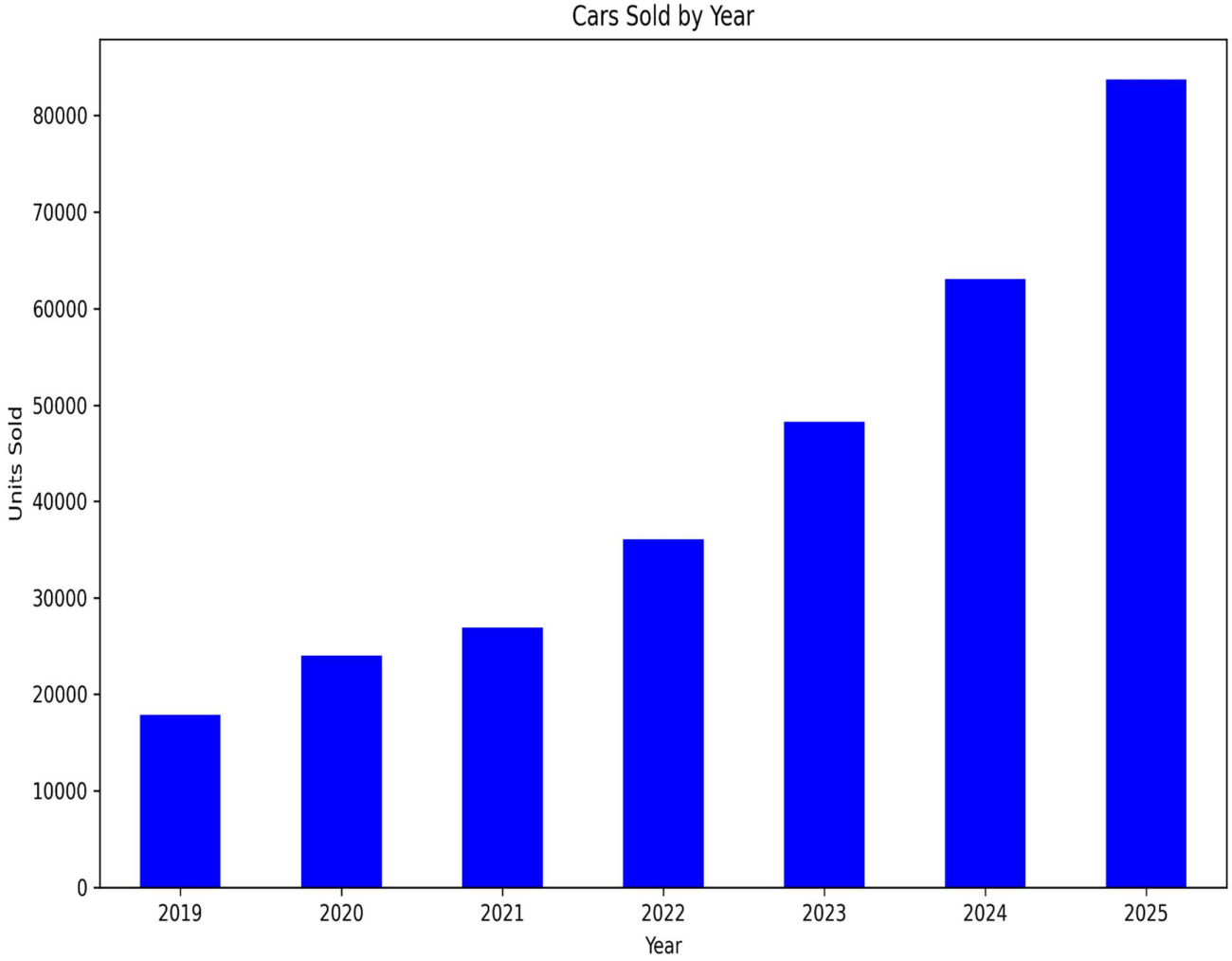
A realistic architecture:

Layer	Purpose	Typical tools
Investigation	Explore the raw dataset first — check shape, column types, missing values, ranges, and distributions before doing any real analysis	`df.info()`, `df.describe()`, `df.head()`, `df.isnull().sum()` in a Jupyter notebook
Stream processing	Not applicable at this scale — there's no continuous flow of events; the equivalent step is grouping/rolling up records after they're already loaded	`pandas.groupby()`, `pivot_table()` used in place of real-time stream aggregation
Storage	Hold the dataset in a form you can repeatedly query and share	Flat files — `toyota.csv` / `toyota_300k_dataset.xlsx`
Batch aggregation	Compute summary metrics — daily/monthly sales totals, average rating by model, profit by region — run manually or via a notebook cell rather than a nightly scheduled job	Jupyter notebook cells using `pandas` (`groupby`, `resample`, `.mean()`, `.sum()`)
Serving/BI layer	Present the aggregated results as charts/tables for reviewPresent the aggregated results as charts/tables for review	`matplotlib` bar charts inside the notebook, or exported to Excel/PDF

Best selling products



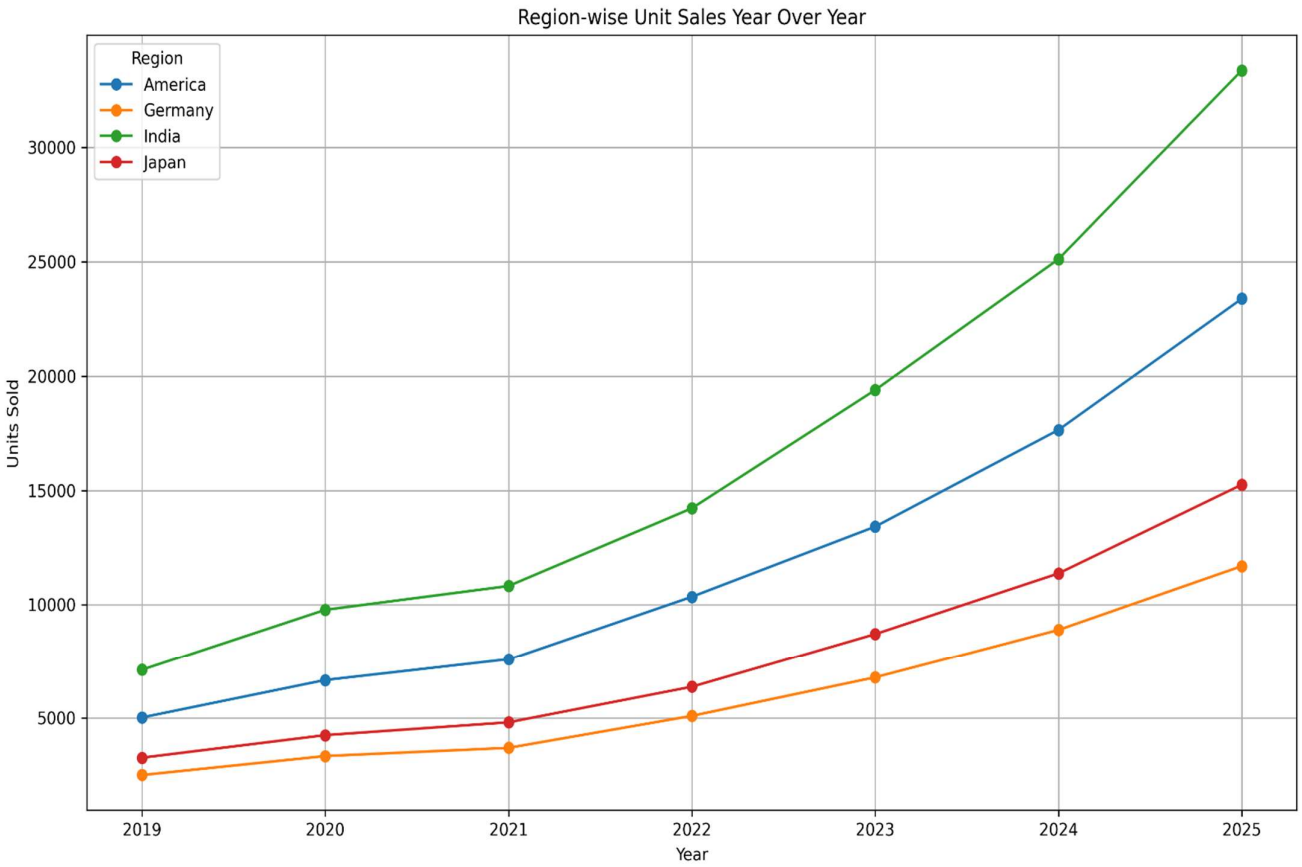
The chart shows that **Etios is the best-selling Toyota model with approximately 40,944 units sold**, followed closely by **Corolla with around 40,354 units**. **Hilux** and **Prius** are also strong performers, with approximately **31,445 and 30,678 units** sold respectively. The top four models together account for around **143,421 units**, showing that a large portion of sales is concentrated among these models. After the top four, sales decline significantly, with **Urban Cruiser selling about 20,295 units**, followed by Land Cruiser at **17,168 units** and Supra at **16,991 units**. At the lower end, **Vitz has the lowest sales at approximately 2,065 units**, while Glanza, Avalon and C-HR each sell around **4,000 units**. Overall, the data indicates that **Etios and Corolla are the strongest-selling products**, while Toyota may need to focus more marketing and product strategies on lower-performing models to improve their sales.



The chart shows a **strong and consistent upward trend in Toyota car sales from 2019 to 2025**. In **2019, approximately 17,731 cars** were sold, which increased to around **23,950 units in 2020**, representing a growth of about **35.1%**. In 2021, sales reached approximately **26,807 units**, an increase of **11.9%** over 2020. Sales then accelerated in 2022 to around **35,966 units**, giving a significant **34.2% year-on-year growth**. In 2023, Toyota sold approximately **48,235 units**, which was an increase of **34.1%** compared with 2022. The growth became even stronger in 2024, when sales reached approximately **63,025 units**, representing **30.7% growth**. Finally, in **2025, sales reached approximately 83,782 units**, the highest level in the entire period, with a year-on-year increase of about **32.9%**.

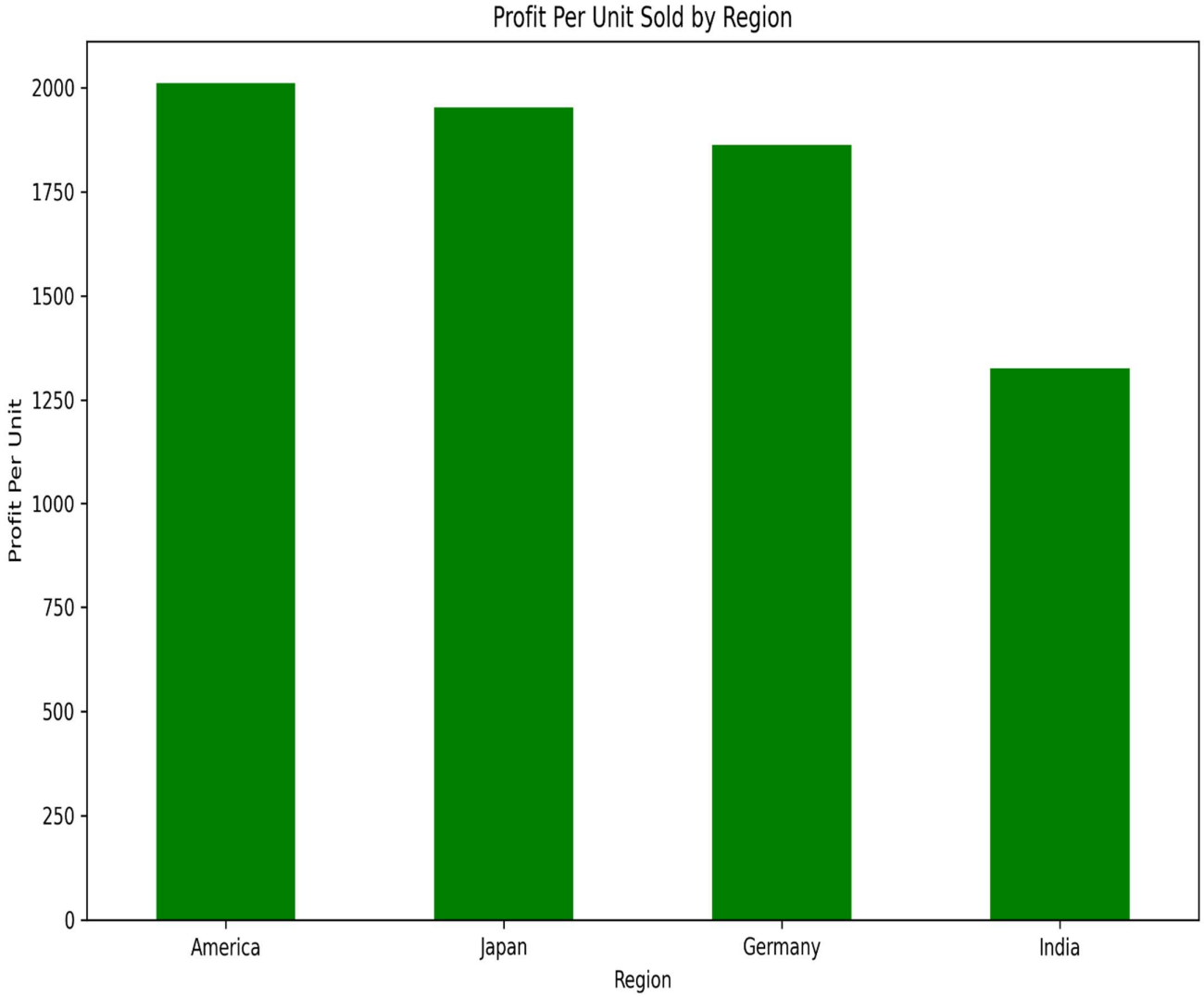
Toyota's annual sales increased from approximately **17,731 units in 2019 to 83,782 units in 2025**, an absolute increase of about **66,051 units**. This means sales became approximately **4.73 times higher** over the seven-year period, equivalent to an overall growth of approximately **372.5%**.

Regional demand

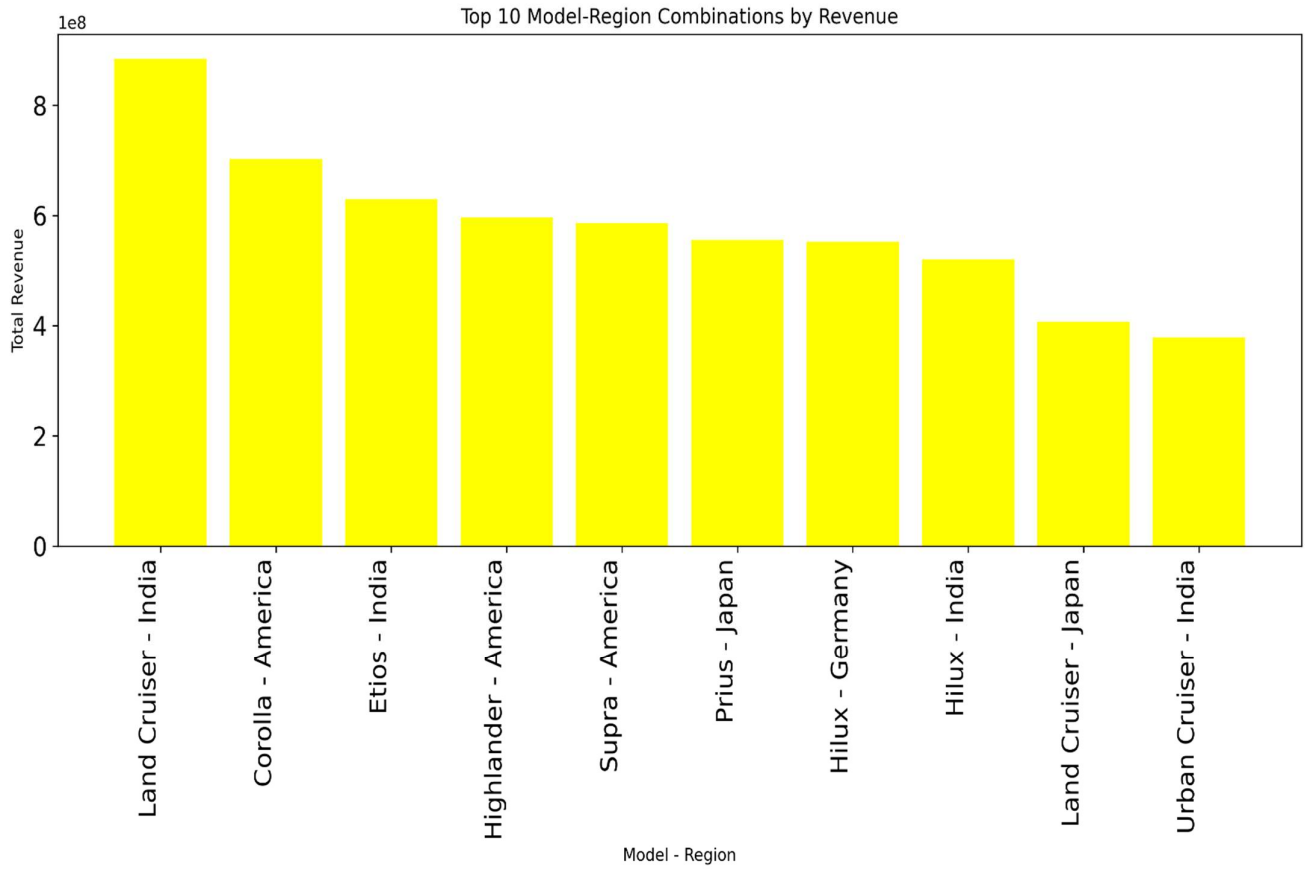


The chart shows a **strong year-over-year increase in sales across all four regions from 2019 to 2025**. **India is the leading market**, increasing from approximately **7,100 units in 2019 to 33,300 units in 2025**, an increase of **26,200 units or about 369%**. India also recorded the highest total sales over the seven years, with approximately **119,700 units**. **America** increased from **5,000 units to 23,400 units**, gaining **18,400 units or 368%**, with total sales of approximately **84,000 units**. **Japan** grew from **3,200 units in 2019 to 15,200 units in 2025**, an increase of **12,000 units or 375%**, making it the region with the highest percentage growth. **Germany** increased from **2,500 units to 11,700 units**, adding **9,200 units or 368%**, with approximately **42,000 total units** sold during the period.

In terms of **combined regional sales**, the four regions sold approximately **17,800 units in 2019**, increasing to **24,000 in 2020**, **26,900 in 2021**, **36,000 in 2022**, **48,300 in 2023**, **62,900 in 2024**, and **83,600 units in 2025**. Across all four regions, the estimated seven-year total is approximately **299,300 units**.

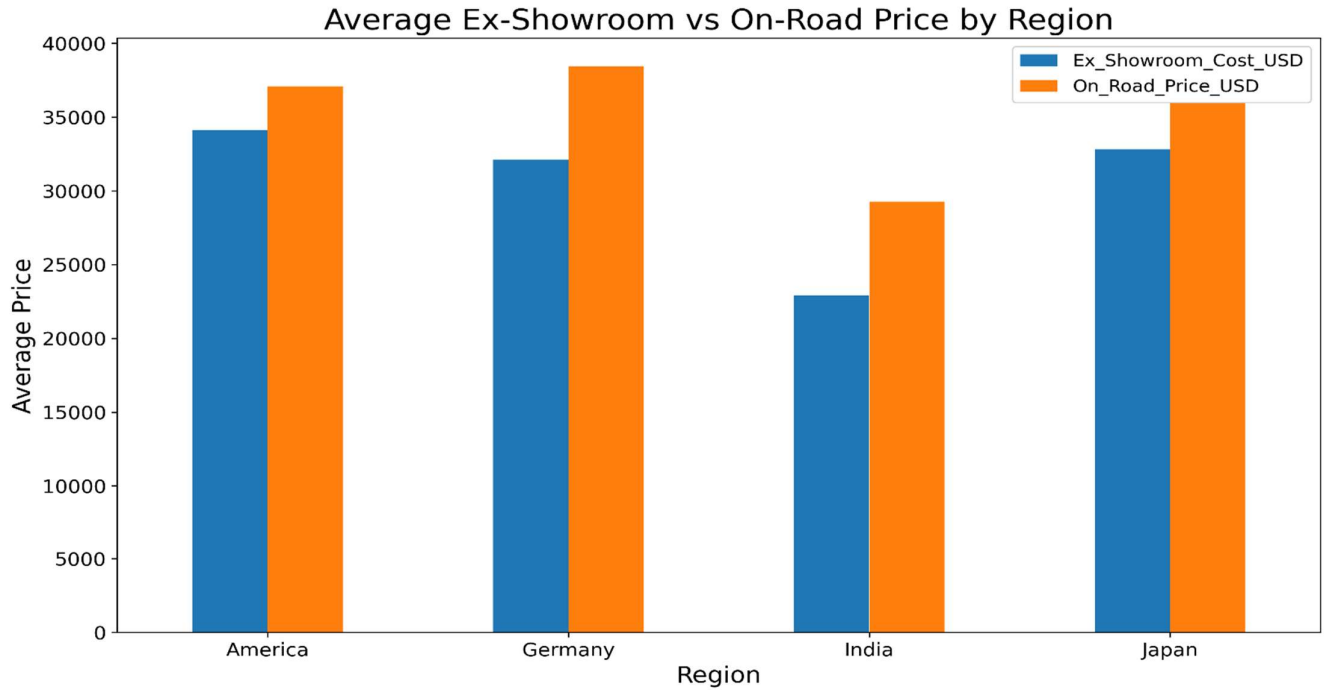


The chart shows that **America has the highest profit per unit at approximately ₹2,010 per car**, followed by **Japan at around ₹1,950**, **Germany at approximately ₹1,860**, and **India at about ₹1,325 per unit**. America therefore earns about **₹685 more profit per car than India**, which is approximately **52% higher**. Japan's profit per unit is around **₹625 higher than India**, while Germany earns approximately **₹535 more per car** than India. The difference between the highest and lowest regions is approximately **₹685 per unit**. Overall, the average profit per unit across the four regions is approximately **₹1,786**, with America performing about **12.5% above the average** and India about **25.8% below the average**. This indicates that although India has the highest sales volume in the regional analysis, **America provides the strongest profit per vehicle**, suggesting better pricing, margins, or product mix in the American market.



The chart shows the **Top 10 Model–Region combinations by revenue**, with revenue values shown in **1e8 scale**, meaning each unit on the vertical axis represents **₹100 million (₹10 crore)**. The strongest combination is **Land Cruiser – India**, generating approximately **₹885 million (₹88.5 crore)** in revenue. It is followed by **Corolla – America** at around **₹702 million (₹70.2 crore)** and **Etios – India** at approximately **₹628 million (₹62.8 crore)**. **Highlander – America** generated about **₹595 million (₹59.5 crore)**, while **Supra – America** generated approximately **₹583 million (₹58.3 crore)**.

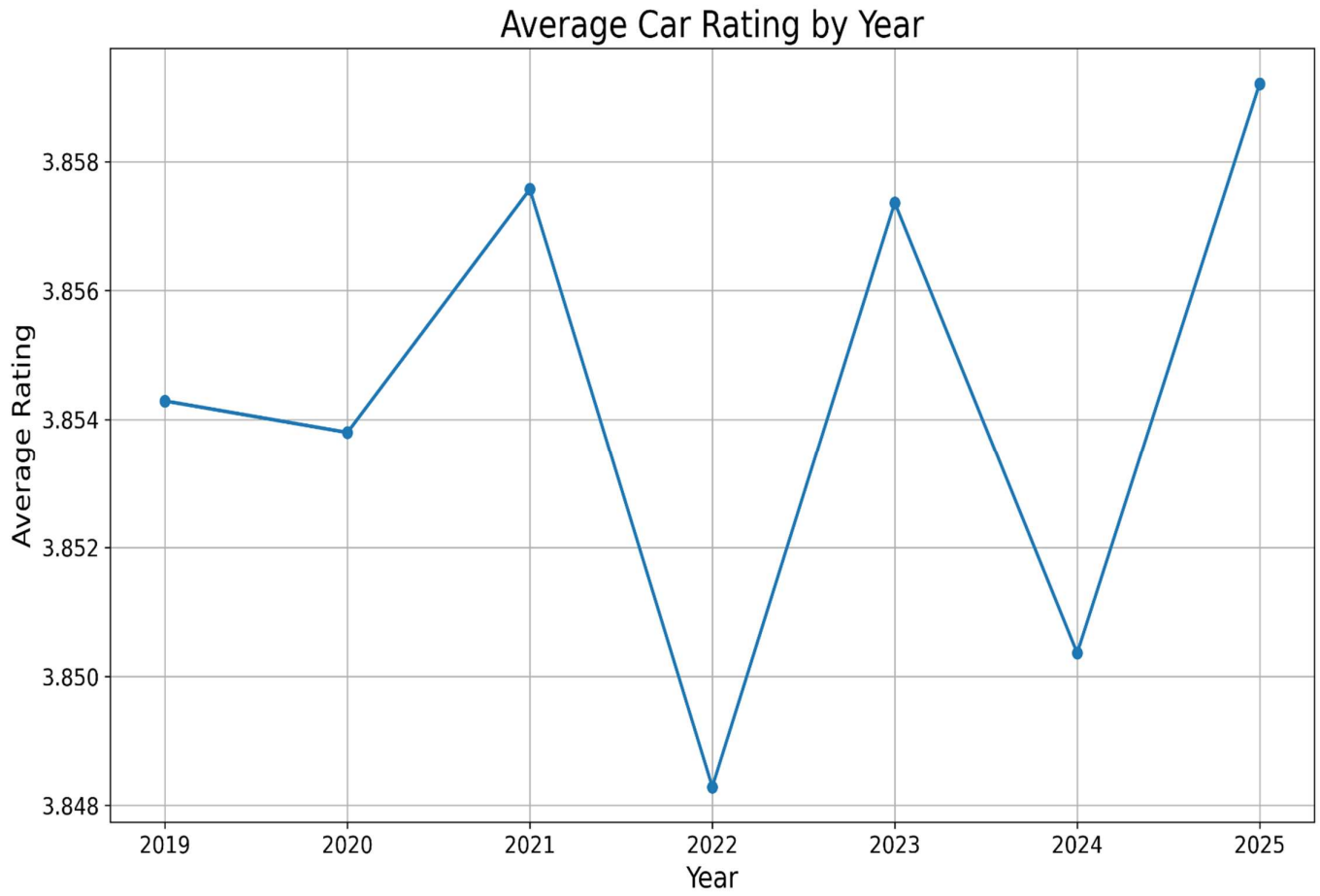
The next major combinations are **Prius – Japan at ₹554 million (₹55.4 crore)**, **Hilux – Germany at ₹550 million (₹55.0 crore)**, and **Hilux – India at approximately ₹519 million (₹51.9 crore)**. **Land Cruiser – Japan** generated around **₹405 million (₹40.5 crore)**, while **Urban Cruiser – India** was tenth with approximately **₹376 million (₹37.6 crore)**.





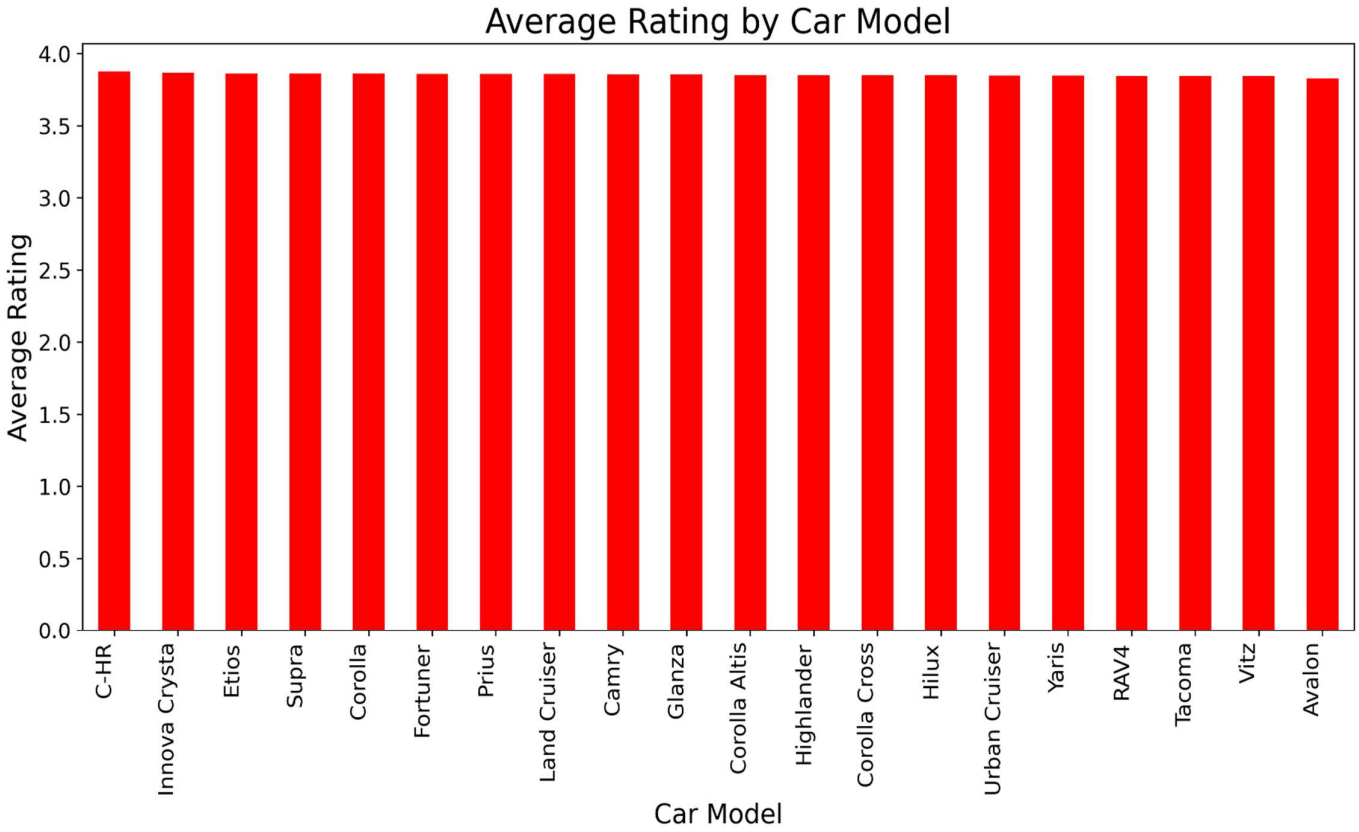
The graph shows that Germany has the highest average on-road price, at approximately **$38,500**, followed by Japan at around **$38,000** and America at about **$37,000**. India has the lowest average on-road price, at approximately **$29,000**.The gap between ex-showroom and on-road prices varies significantly by region. India has one of the largest gaps, at approximately **$6,350**, meaning taxes, registration, and other additional charges add a substantial amount to the base price. Germany has a very similar gap of around **$6,300**. Japan has a moderate gap of approximately **$3,200**, while America has the lowest gap at around **$3,000**.

Rating distribution



The chart shows that the **average car rating remained very stable between 2019 and 2025**, staying within a narrow range of approximately **3.848 to 3.859 out of 5**. In **2019**, the average rating was about **3.8543**, which slightly decreased to **3.8538 in 2020**, a fall of approximately **0.0005 points (0.01%)**. The rating then increased to **3.8576 in 2021**, an improvement of about **0.0038 points**. However, **2022 recorded the lowest average rating at approximately 3.8483**, representing a decrease of around **0.0093 points or 0.24%** compared with 2021. The rating recovered strongly in **2023 to approximately 3.8574**, an increase of about **0.0091 points (0.24%)**. In **2024**, it again declined to approximately **3.8504**, a reduction of about **0.0070 points**, before reaching the **highest rating of approximately 3.8592 in 2025**.

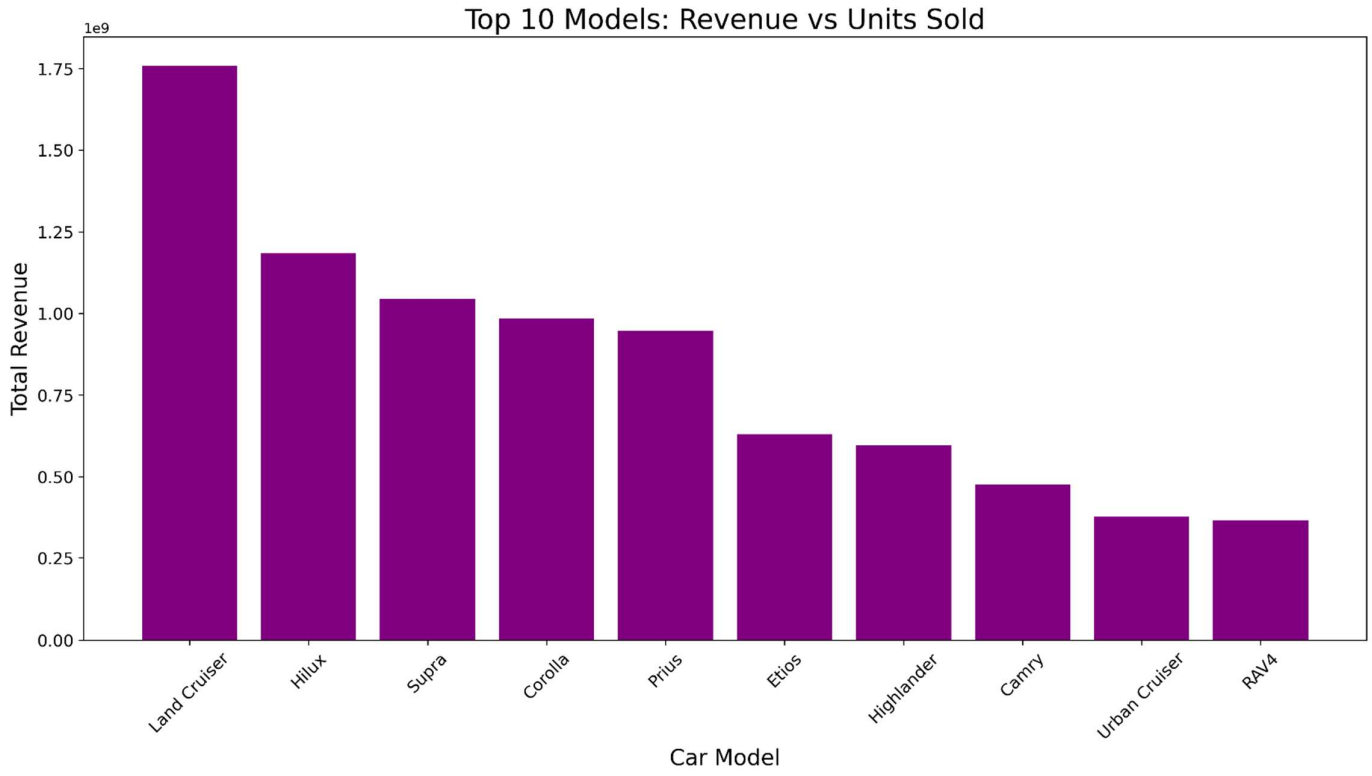
The difference between the **highest rating in 2025 (3.8592)** and the **lowest rating in 2022 (3.8483)** is only about **0.0109 rating points**, showing that customer ratings were highly consistent throughout the seven-year period.



The chart shows that the **average ratings of all 20 Toyota car models are relatively close to each other**, indicating consistent customer satisfaction across the product range. **C-HR has the highest average rating at approximately 3.72/5**, followed by **Innova Crysta at about 3.72**, while **Etios, Supra and Corolla are around 3.71**. Fortuner, Prius and Land Cruiser are approximately **3.70**, while Camry and Glanza are around **3.70**. At the lower end, **Avalon has the lowest average rating at approximately 3.67/5**. The difference between the highest-rated model, C-HR (**3.72**), and the lowest-rated model, Avalon (**3.67**), is only about **0.05 rating points**, showing that there is **very little variation between models**.

KPI	Value
Average Product Rating	**3.85/5**
Highest Model Rating	**~3.88/5**
Lowest Model Rating	**~3.83/5**
Rating Range	**~0.05**
Rating Scale	**5**

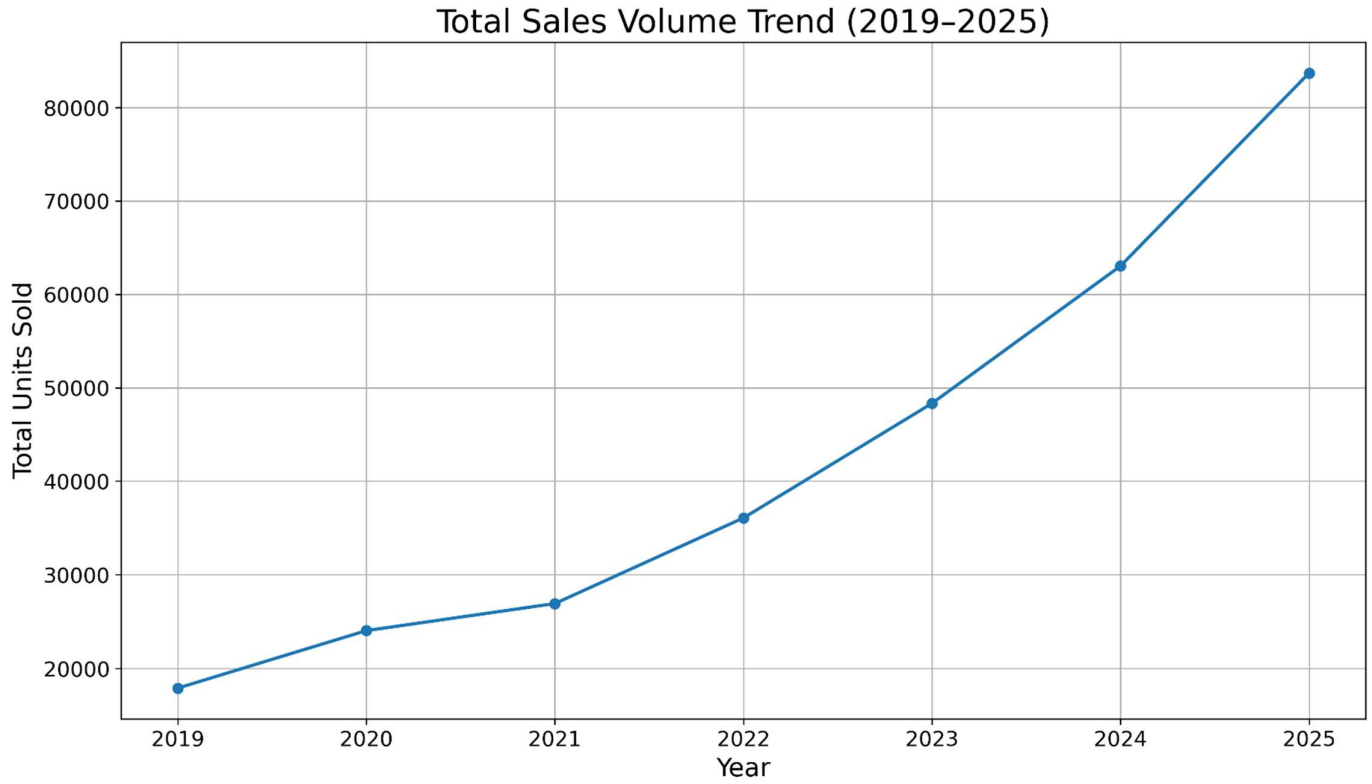
Revenue and profit



The chart shows the **Top 10 Toyota models by total revenue**, with **Land Cruiser clearly leading at approximately ₹1.76 billion (₹176 crore)**. **Hilux** ranks second with around **₹1.18 billion (₹118 crore)**, followed by **Supra at ₹1.04 billion (₹104 crore)**, **Corolla at ₹0.98 billion (₹98 crore)**, and **Prius at approximately ₹0.95 billion (₹95 crore)**. Etios generated about **₹0.63 billion (₹63 crore)**, Highlander around **₹0.60 billion (₹60 crore)**, Camry approximately **₹0.48 billion (₹48 crore)**, Urban Cruiser around **₹0.38 billion (₹38 crore)**, and RAV4 approximately **₹0.36 billion (₹36 crore)**.

KPI	Value
Total Revenue (2019–2025)	**$10.91 Billion**
Average Annual Revenue	**$1.56 Billion**
Highest Annual Revenue	**$3.05 Billion**
Highest Revenue Year	**2025**
Highest Revenue Model	**Land Cruiser**
Land Cruiser Total Revenue	**~$1.76 Billion**

Daily sales trend



The chart shows a **strong and continuous increase in Toyota's total sales volume from 2019 to 2025**. Sales increased from approximately **17,800 units in 2019** to **24,000 units in 2020**, an increase of about **6,200 units or 34.8%**. In 2021, sales reached approximately **26,900 units**, adding **2,900 units or 12.1%**. The growth accelerated in 2022, when sales reached around **36,000 units**, an increase of approximately **9,100 units or 33.8%**. In 2023, sales increased further to about **48,300 units**, adding **12,300 units or 34.2%**. Sales then reached approximately **63,000 units in 2024**, representing an increase of **14,700 units or 30.4%**. The strongest absolute growth occurred in **2025**, when sales reached approximately **83,700 units**, an increase of about **20,700 units or 32.9%** compared with 2024.

Across the full **2019–2025 period**, total sales increased by approximately **65,900 units**, from 17,800 to 83,700 units. This represents an overall growth of approximately **370%**, meaning annual sales in 2025 were about **4.7 times** the 2019 level.

Summary of Key matrics

Metric	Value (simulated)
Total units sold (7 years)	**~300k**
Total revenue (7 years)	**~$11 billion**
Average product rating	**3.85**
Top single-day seller	**400 units**
Regions tracked	**4**
Top seller model	**Land cruiser**

Next Steps to Move From Descriptive → Predictive

This report currently explains what happened in Toyota's sales data from 2019 to 2025, such as which models sold the most units, which regions generated the highest revenue, changes in customer ratings, and overall sales growth. The next step is to use predictive analytics to understand what is likely to happen in the future.

- **Sales forecasting**: Predict future Toyota sales by model, region, and year using historical sales trends.
- **Regional sales prediction**: Forecast future sales for America, Germany, India, and Japan to identify which region is likely to grow the fastest.
- **Revenue prediction**: Use previous revenue, sales volume, price, and discount data to estimate future revenue.
- **Rating prediction**: Analyze previous customer ratings to identify whether the average rating of a model is likely to improve or decline.

Key findings

Volume vs. Revenue Leadership ● The Etios (41,003 units) and Corolla (40,412 units) are the two best-selling models by volume, but the Land Cruiser is the single largest revenue contributor ($1.76 billion) despite selling fewer than half as many units as the Etios — the highest-revenue and highest-volume models are not the same vehicle.

Sustained, Uninterrupted Growth ● Total unit sales rose every year from 2019 (17,912 units) to 2025 (83,700 units), a ~29% compound annual growth rate with zero years of decline; 2025 is the single best-selling year in the dataset.

Stable Customer Satisfaction ● The average customer rating (3.85/5) has remained essentially flat across all 84 months of the dataset, moving in a narrow 3.75–3.95 band with no sustained downward trend; both the highest- and lowest-rated months occurred in 2019, not near the end of the series.

India Leads on Volume, Not on Margin ● India is the largest market by unit sales (119,833 units, ~40% of global volume) in every single year of the dataset — a stable #1 ranking — but it is also the least profitable region per unit sold ($1,326 vs. $2,011 in America) and carries the heaviest tax/registration load (27.7% of ex-showroom cost, roughly three times America's 8.7%).

America Is the Profitability Leader ● America generates the highest average profit per unit ($2,011), narrowly ahead of Japan ($1,952) and Germany ($1,862), despite having a smaller unit volume than India.

Model Ratings Are Tightly Clustered ● Average rating by model spans a narrow range from 3.828 (Avalon, lowest) to 3.876 (C-HR, highest) — a difference of less than 0.05 points across all 20 models — indicating no single model has a meaningful quality or satisfaction problem relative to the rest of the lineup.

One Combination Dominates Revenue ● Land Cruiser sales within India alone generated $884.8 million — more than half of the model's total worldwide revenue — making Land Cruiser × India the single largest model region revenue driver in the dataset, ahead of Corolla × America ($702.4M) and Etios × India ($630.1M).

Discounting Is the Exception, Not the Norm ● 35.2% of all transactions carry zero discount, and the median discount is only $122 against a mean of $377 — most customers pay close to the listed on-road price, while a minority receive substantially larger discounts, pulling the average upward.

Reccomendation:-

Consistent with the descriptive scope of this report, the following are brief, evidence-based observations drawn directly from the findings above — not a detailed predictive strategy.

● Protect and grow margin in India rather than volume alone: since India already leads on units but trails on profit per unit, initiatives that improve regional margin (e.g. reviewing discount depth, or trim/option mix) may add more bottom-line value than further volume growth in a market that is already #1.

● Treat the Land Cruiser × India pairing as a flagship revenue relationship worth actively protecting — inventory availability, financing options, and after-sales service capacity for this specific combination disproportionately affect total company revenue.

● Continue monitoring rating trend monthly, even though no decline was found: because the current pattern is flat rather than improving, small monitoring dashboards (a rolling 6-month average, as used in Figure 3) would allow an early, genuine downturn to be caught quickly rather than dismissed as noise.

● Use America and Japan's higher per-unit profitability as a benchmark when reviewing pricing and discount policy in Germany and India, to understand which combination of tax structure, discount depth, and model mix explains the profitability gap.

● Investigate the Avalon's comparatively lower average rating (3.828, the lowest of all 20 models) even though the gap versus the rest of the range is small, since it is the one model whose satisfaction score is a consistent outlier at the bottom of the ranking.

Appendix:-

Full Units-Sold-by-Model Table

Model	Units Sold
Etios	41,003
Corolla	40,412
Hilux	31,543
Prius	30,735
Urban Cruiser	20,365
Land Cruiser	17,252
Supra	17,087
Yaris	15,592
Highlander	14,413
Camry	14,411
RAV4	10,792
Corolla Cross	10,307
Corolla Altis	5,875
Tacoma	5,748
Innova Crysta	5,062
Fortuner	4,731
C-HR	4,224
Avalon	4,177
Glanza	4,085
Vitz	2,186

Full Average-Rating-by-Model Table

Model	Avg. Rating
C-HR	3.876
Innova Crysta	3.865
Etios	3.861
Supra	3.861
Corolla	3.860
Fortuner	3.859
Prius	3.859
Land Cruiser	3.858
Camry	3.856
Glanza	3.855
Corolla Altis	3.851
Highlander	3.851
Corolla Cross	3.850
Hilux	3.849
Urban Cruiser	3.847
Yaris	3.846
RAV4	3.845
Tacoma	3.844
Vitz	3.843
Avalon	3.828